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Project Title	Multi-Echelon Supply Chain Network Design via Mixed-Integer Linear Programming (MILP)
Week	Week 3

WEEK-3 TASK

Task Name

Multi-Echelon Supply Chain Network Design via Mixed-Integer Linear Programming (MILP)

1. Abstract

This project uses **Mixed-Integer Linear Programming (MILP)** to design an efficient supply chain network. The model includes factories, warehouses, and customer zones. It minimizes transportation and operating costs while satisfying capacity, demand, and service-level constraints.

2. Introduction

A multi-echelon supply chain involves the movement of products through different stages such as **Factory → Warehouse → Customer**. MILP helps determine the optimal transportation and warehouse decisions at minimum cost.

3. Objectives

- Optimize transportation between different levels.
- Satisfy factory production capacity.
- Maintain warehouse storage capacity.
- Meet customer demand.
- Maintain minimum service levels.
- Minimize total supply chain cost.

4. Supply Chain Structure

Factories → Warehouses → Customer Zones

- **Factories:** Produce goods.
- **Warehouses:** Store and distribute goods.
- **Customer Zones:** Receive products according to demand.

5. Decision Variables

- X_{fw} : Quantity transported from factory to warehouse.
- Y_{wc} : Quantity transported from warehouse to customer.
- Z_w : Binary variable indicating warehouse selection.

6. Objective Function

$$\text{Minimize } Z = \sum C_{fw} X_{fw} + \sum C_{wc} Y_{wc}$$

The objective is to minimize total transportation and operating expenditure.

7. Important Constraints

Factory Capacity:

$$\sum X_{fw} \leq \text{Capacity}_f$$

Warehouse Capacity:

$$\sum X_{fw} \leq \text{Capacity}_w$$

Flow Balance:

$$\sum X_{fw} = \sum Y_{wc}$$

Customer Demand:

$$\sum Y_{wc} \geq \text{Demand}_c$$

Minimum Service Level:

$$\sum Y_{wc} \geq \text{ServiceLevel}_c \times \text{Demand}_c$$

Non-Negativity:

$$X_{fw}, Y_{wc} \geq 0$$

8. Constraint Matrix

The MILP model can be represented as:

Minimize CTX

Subject to:

$$\mathbf{AX} < \mathbf{B}$$

$$\mathbf{EX} = \mathbf{D}$$

$$\mathbf{X} > \mathbf{0}$$

Here, **A** represents capacity constraints, **E** represents flow-balance constraints, and **B/D** represent capacity and demand values.

9. Methodology

1. Define factories, warehouses, and customer zones.
2. Collect capacity, demand, and transportation-cost data.
3. Define decision variables.
4. Formulate objective function.
5. Add supply-chain constraints.
6. Solve the MILP model.
7. Analyze the optimal transportation plan.

10. Expected Result

The model provides an **optimal supply chain network** with minimum operating cost while satisfying factory capacity, warehouse capacity, transportation limits, customer demand, and service-level requirements.

11. Applications

- Global logistics
- Warehouse planning
- Transportation optimize
- Distribution management
- Inventory planning
- E-commerce supply chains

12. Conclusion

MILP provides an effective method for optimizing multi-echelon supply chains. It helps organizations **reduce costs, utilize resources efficiently, satisfy customer demand, and improve overall supply chain performance.**